# Basic fMRI, Basic Decoding, & TDT – The Decoding Toolbox

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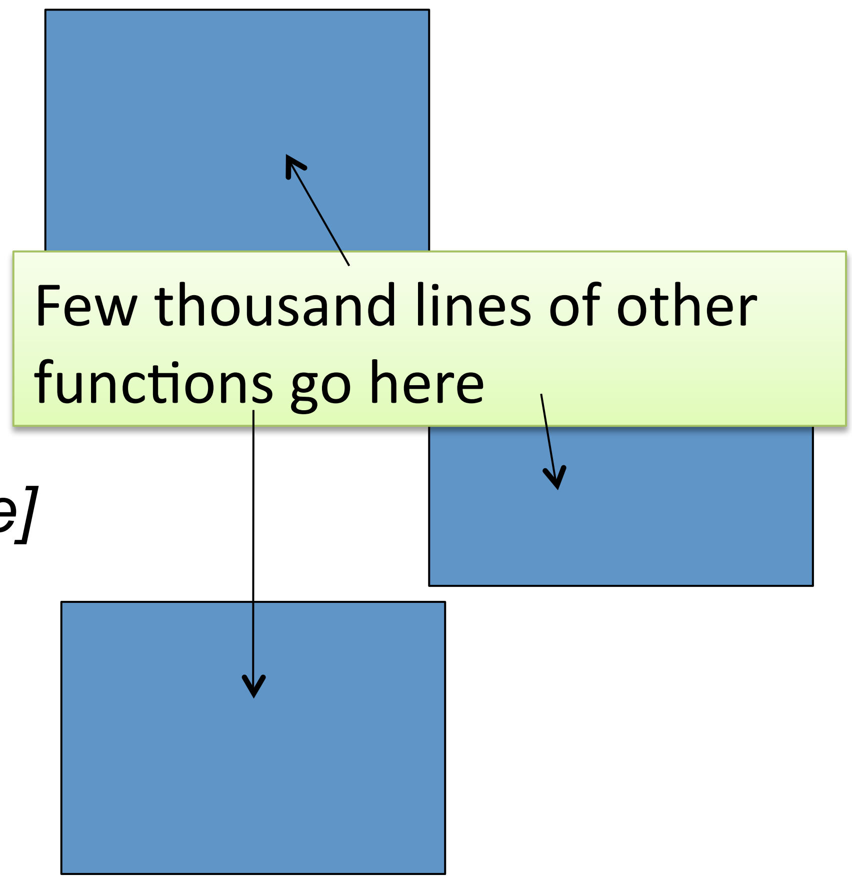
IK2012, March $20^{th}$, 2012

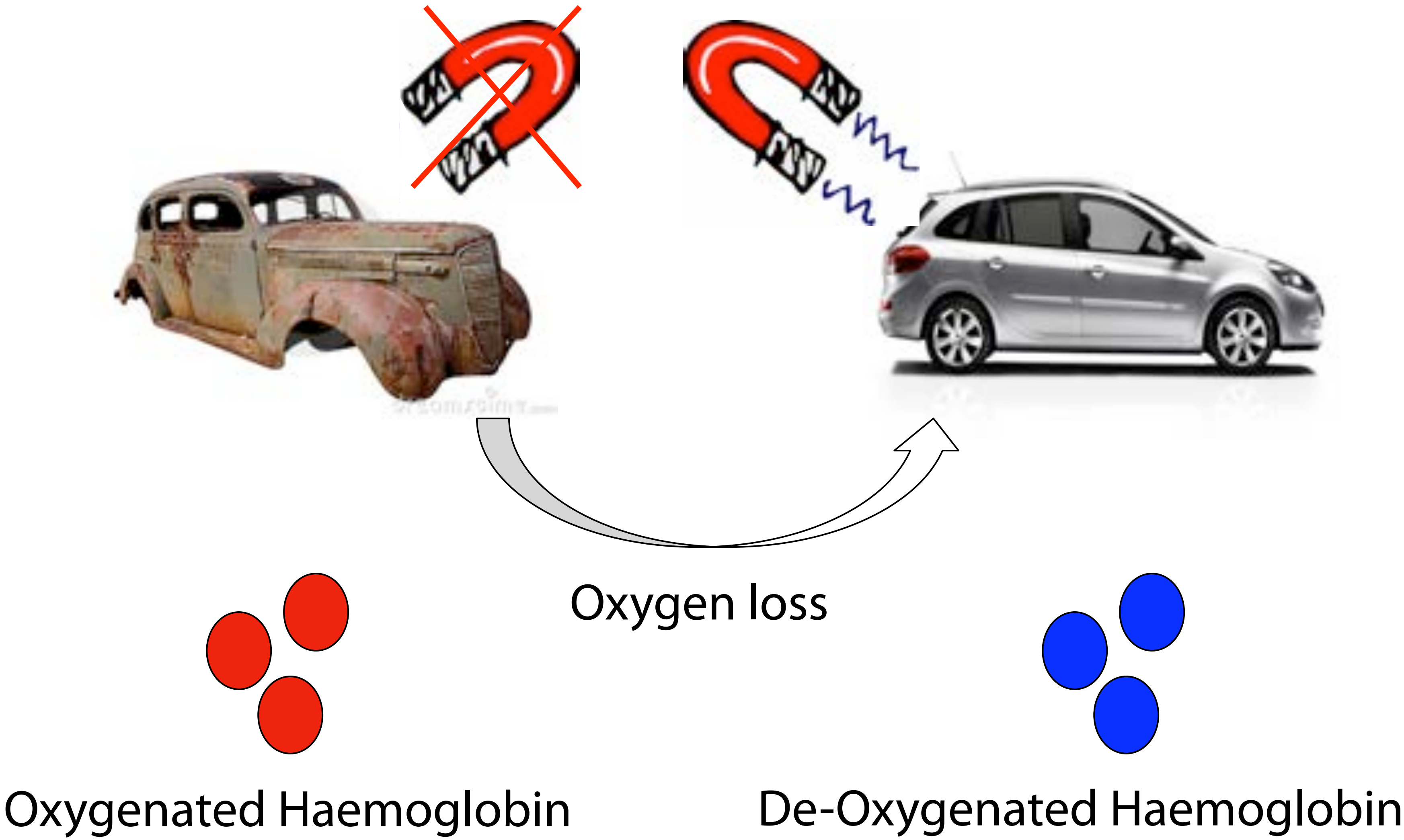
```
for runs = 1:nruns
    ...
    [78 lines of code go here]
    ...
    for set = 1:nsets
     ...
        [258 lines of code go here]
     ...
        output = outputs(1:set)
    end
    % output = outputs(1:set);
end
```



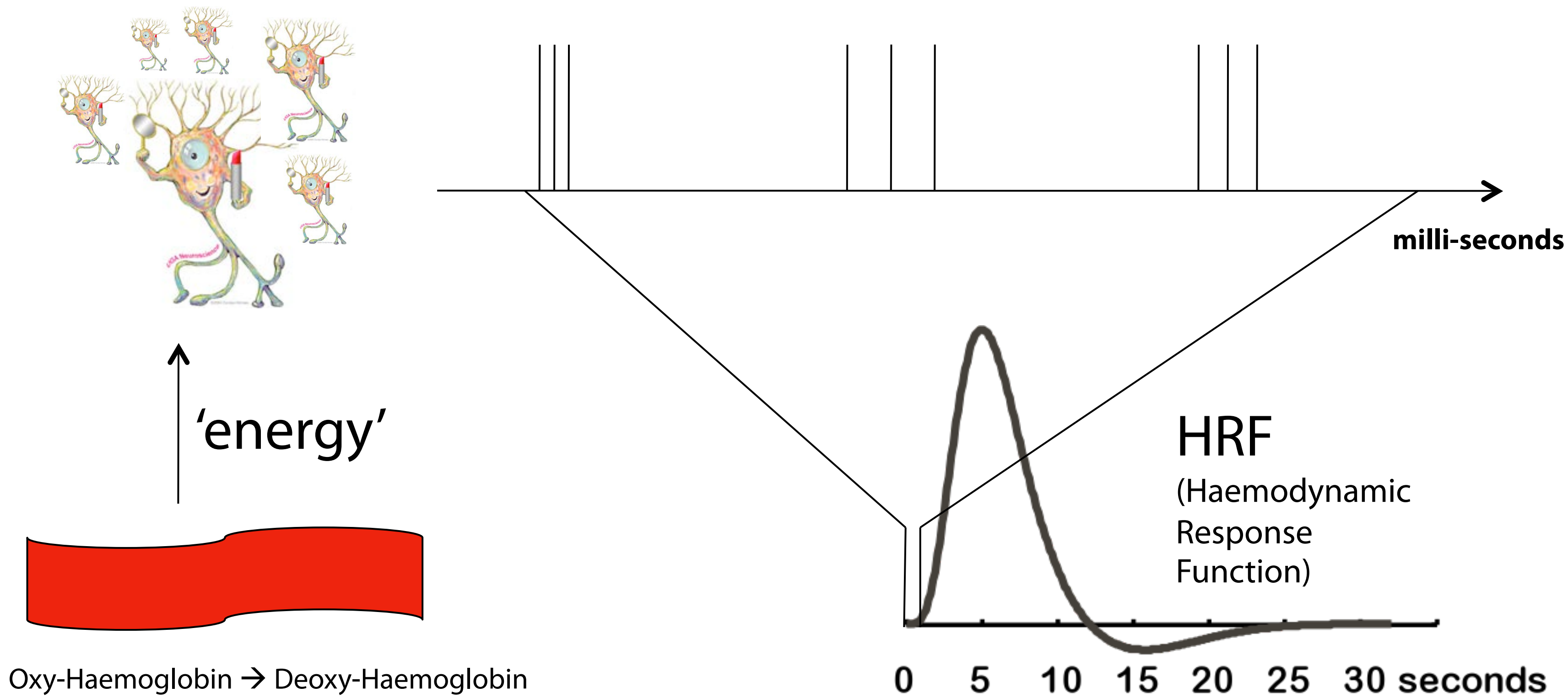
# Outline

- Background
  - Very short: fMRI
  - Decoding
    What and why
  - What to expect
    A Very short overview about The Decoding Toolbox
- Hands-On

# A word on fMRI

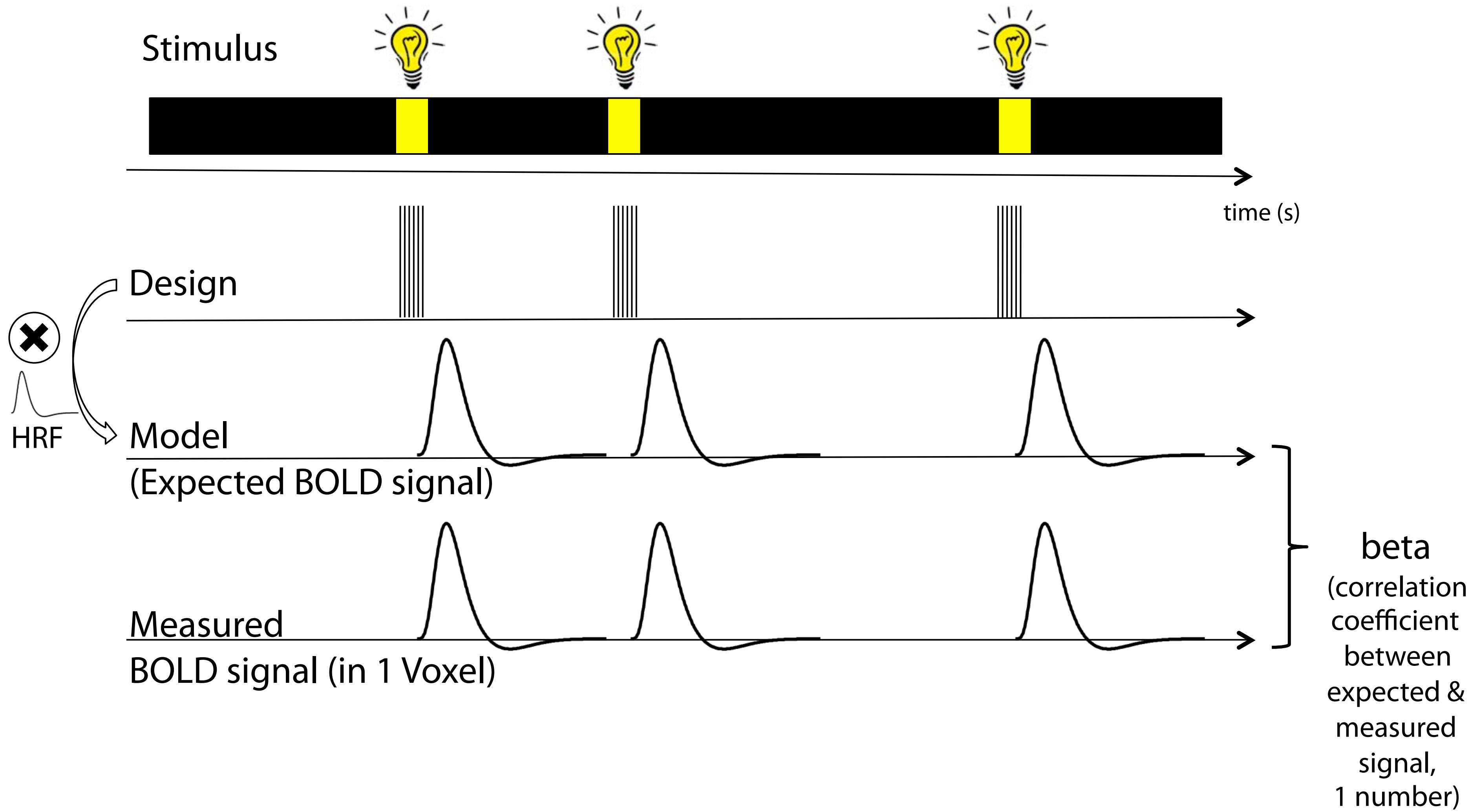


# A word on fMRI

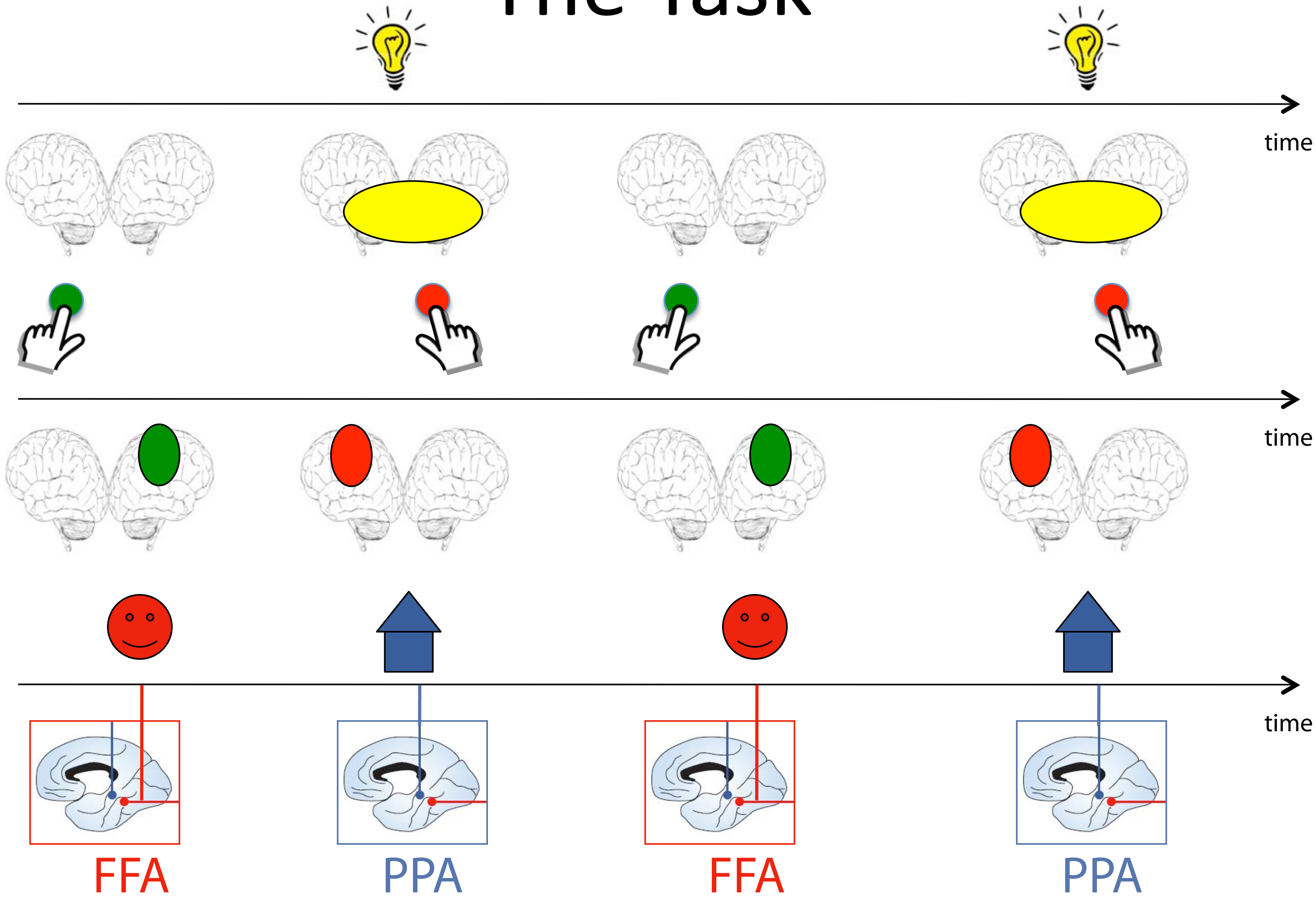



Oxygen unbound ('un'-rost) → Magnetic → 'BOLD' signal

# Modelling BOLD & Betas



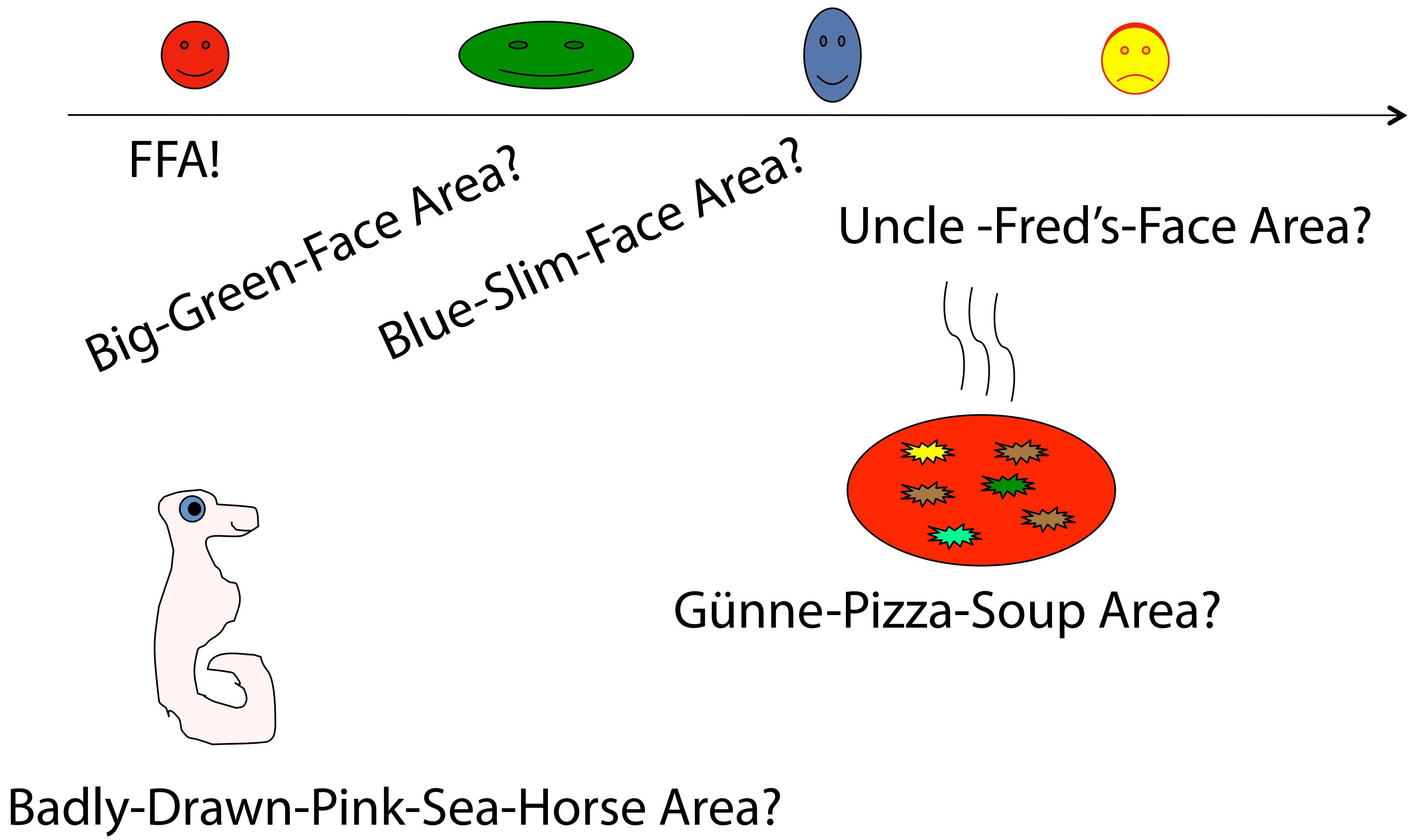
# The Task



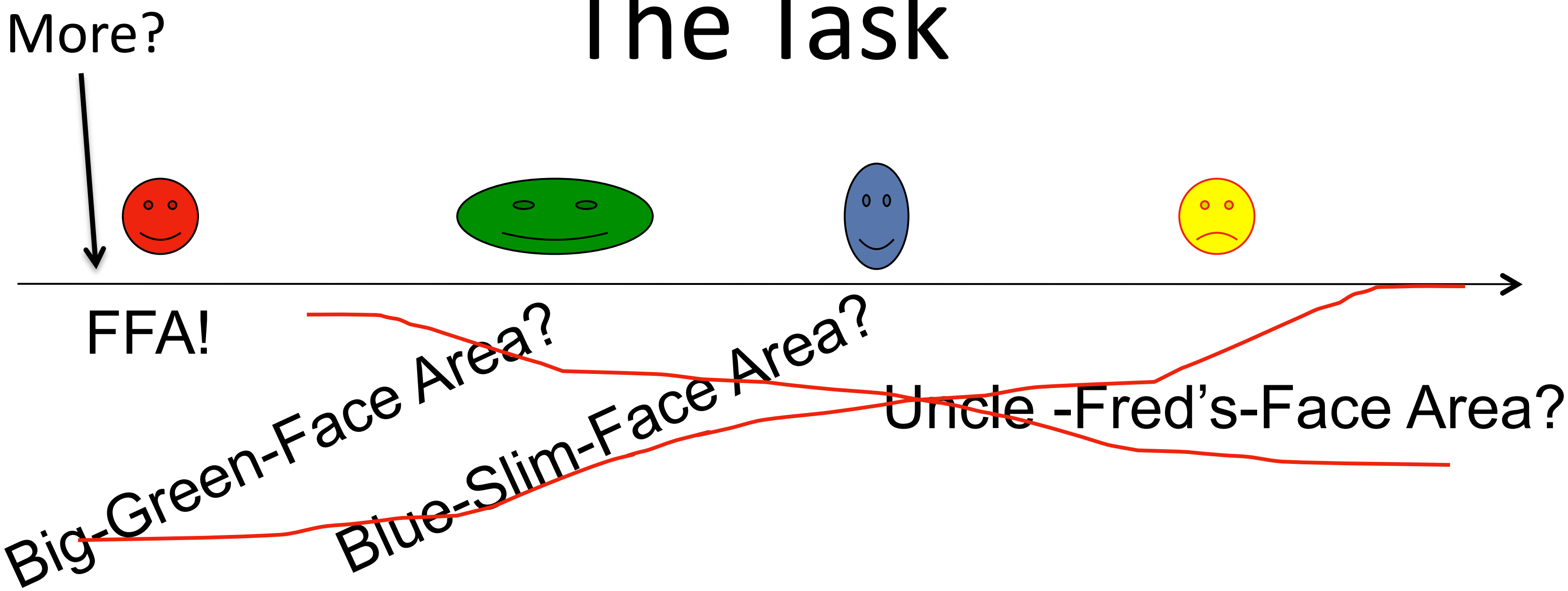

Kanwisher et al, 95, JNeuro

Eppstein et al, 98, Science

# The Task


FFA!
Big-Green-Face Area?
Blue-Slim-Face Area?
Uncle -Fred's-Face Area?
Günne-Pizza-Soup Area?
Badly-Drawn-Pink-Sea-Horse Area?

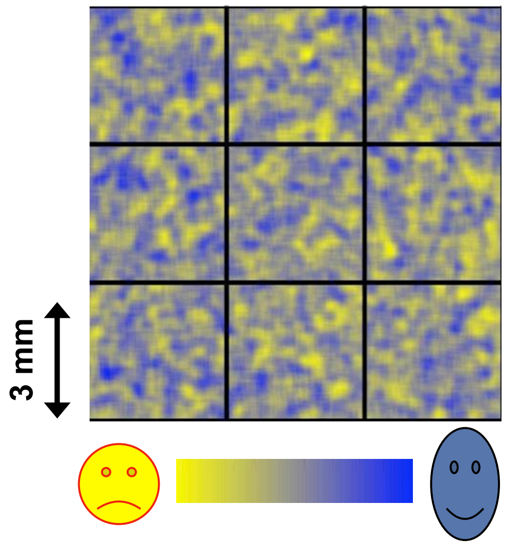
# The Task

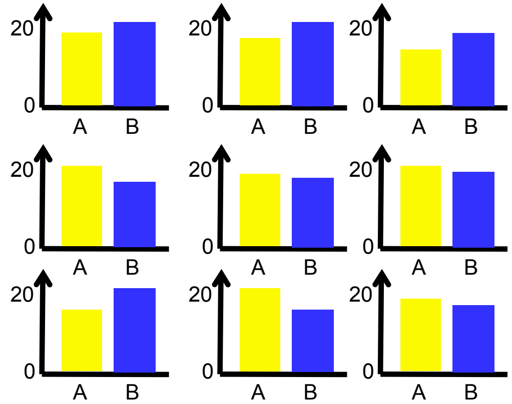



Idea: Different **patterns** of neural activity ***within*** areas encode the difference between different faces (objects/thoughts/…)!

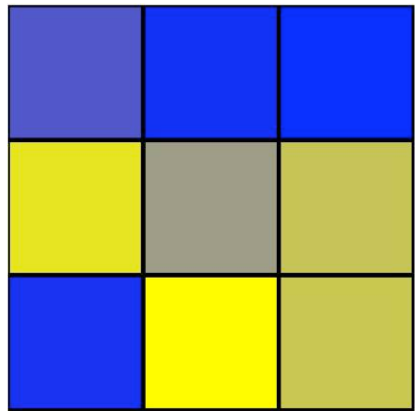
Objective for decoding: Find these areas

# How decoding improves fMRI 'resolution'



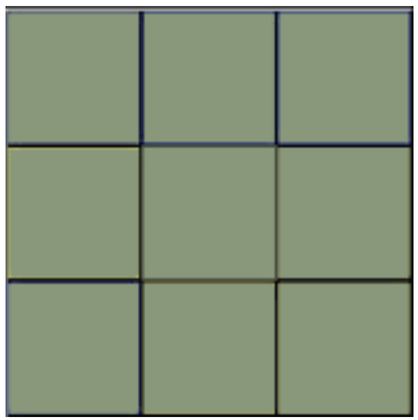




Small differences
in each voxel



Detect difference using
multi-variate decoding methods

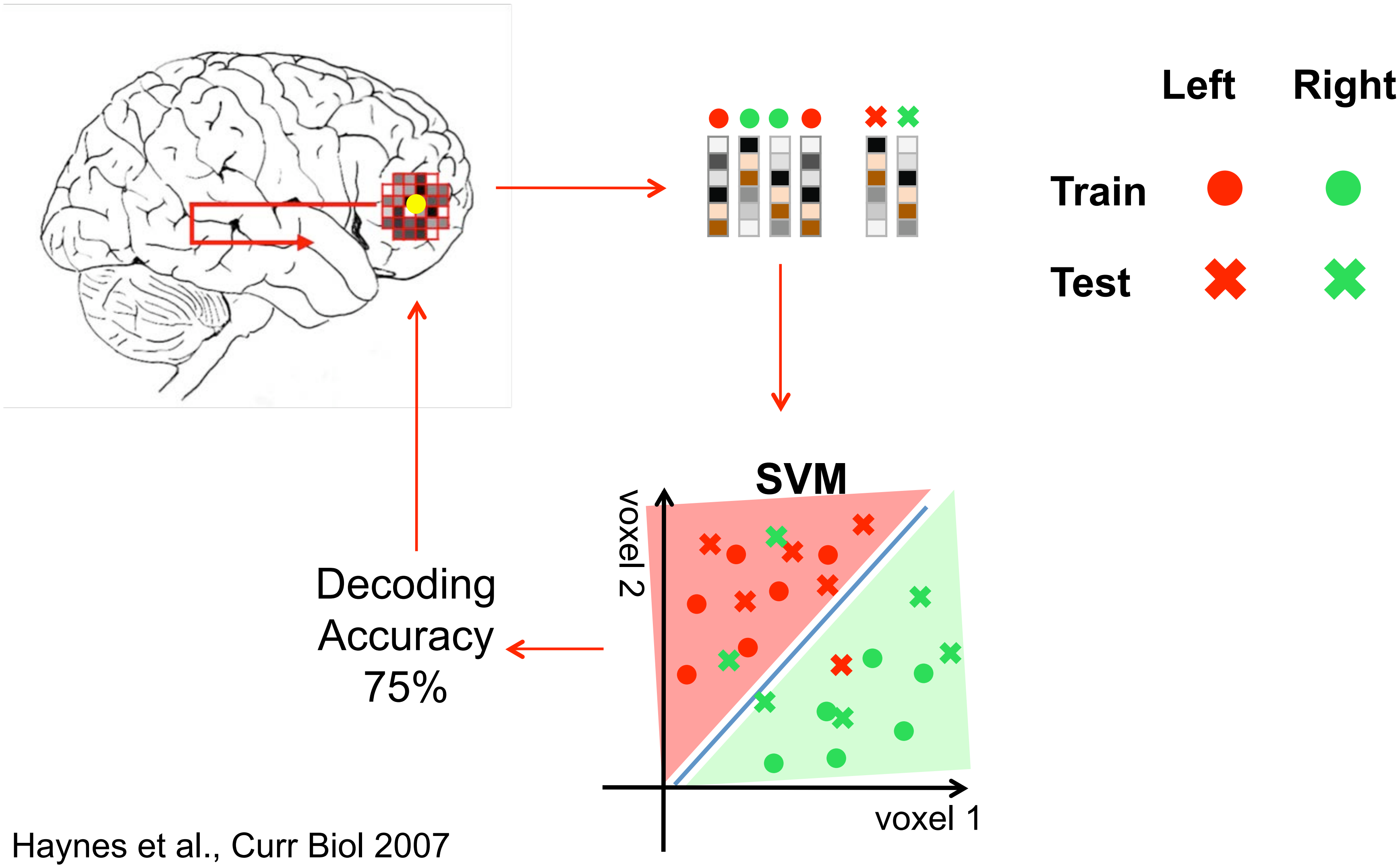
Conventional fMRI:
Smoothing to get rid of noise
→ No difference



# Early decoding studies

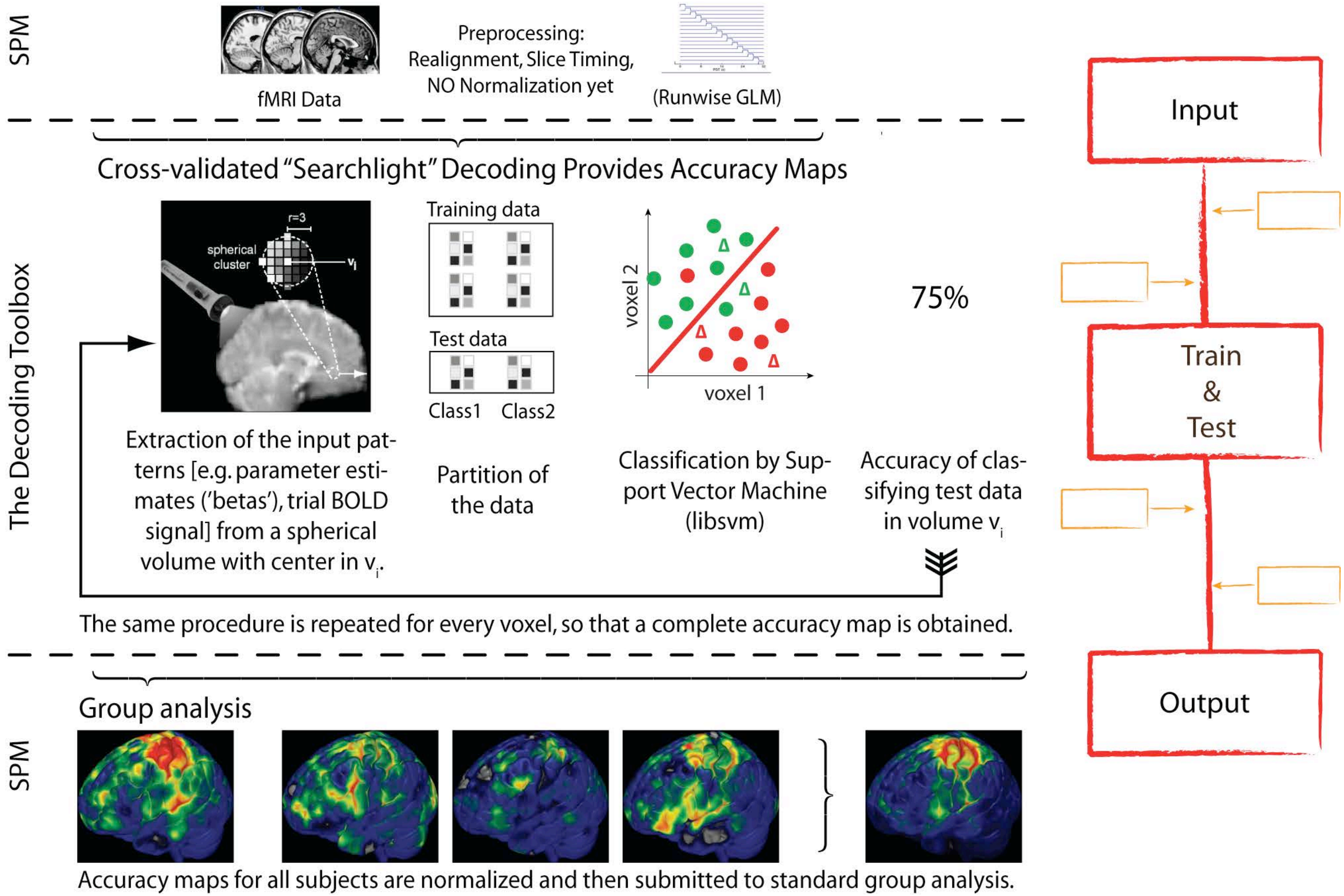
- Haxby
- Kamatami & Tong
- Haynes & Rees

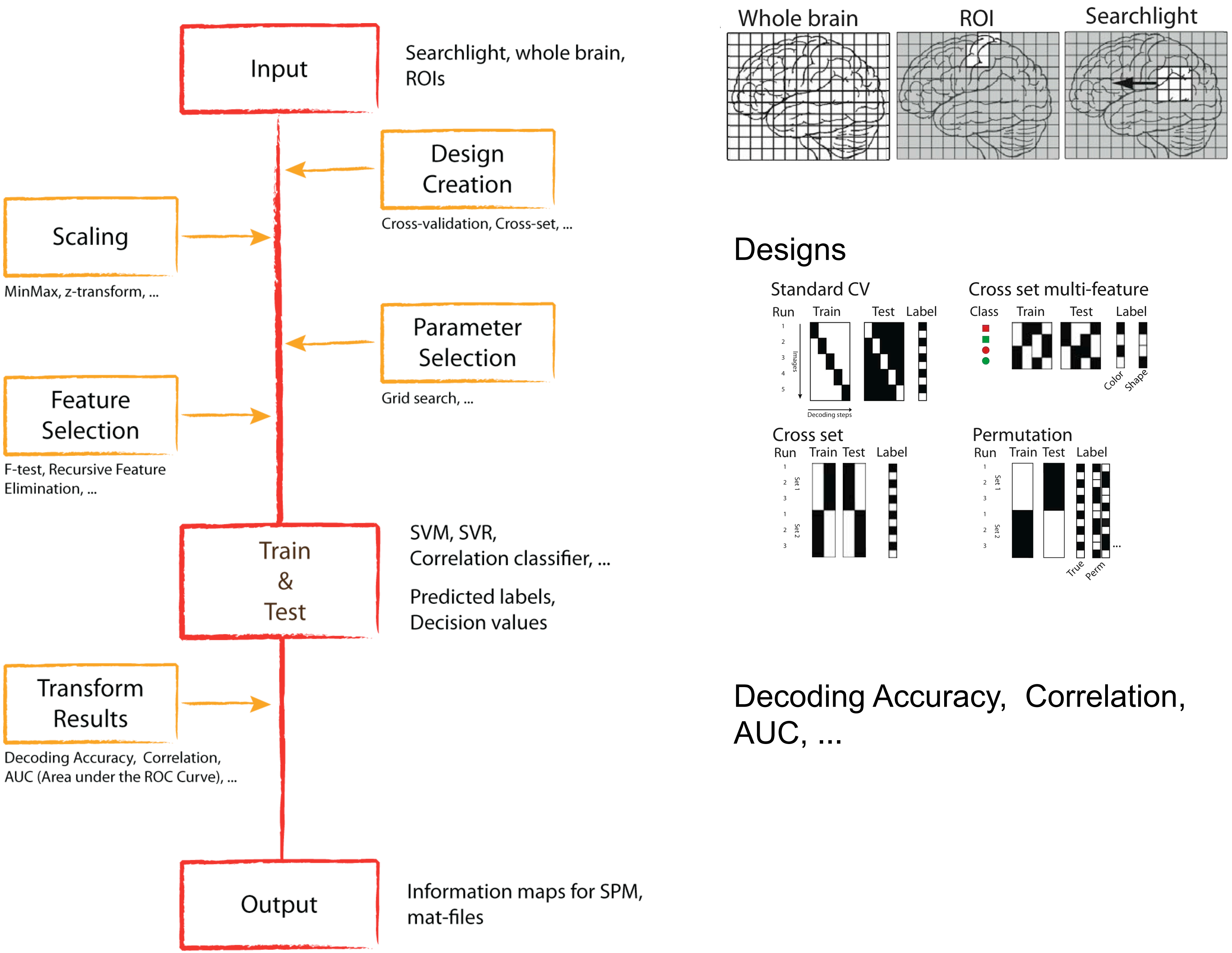
# Searchlight-Decoding




Haynes et al., Curr Biol 2007

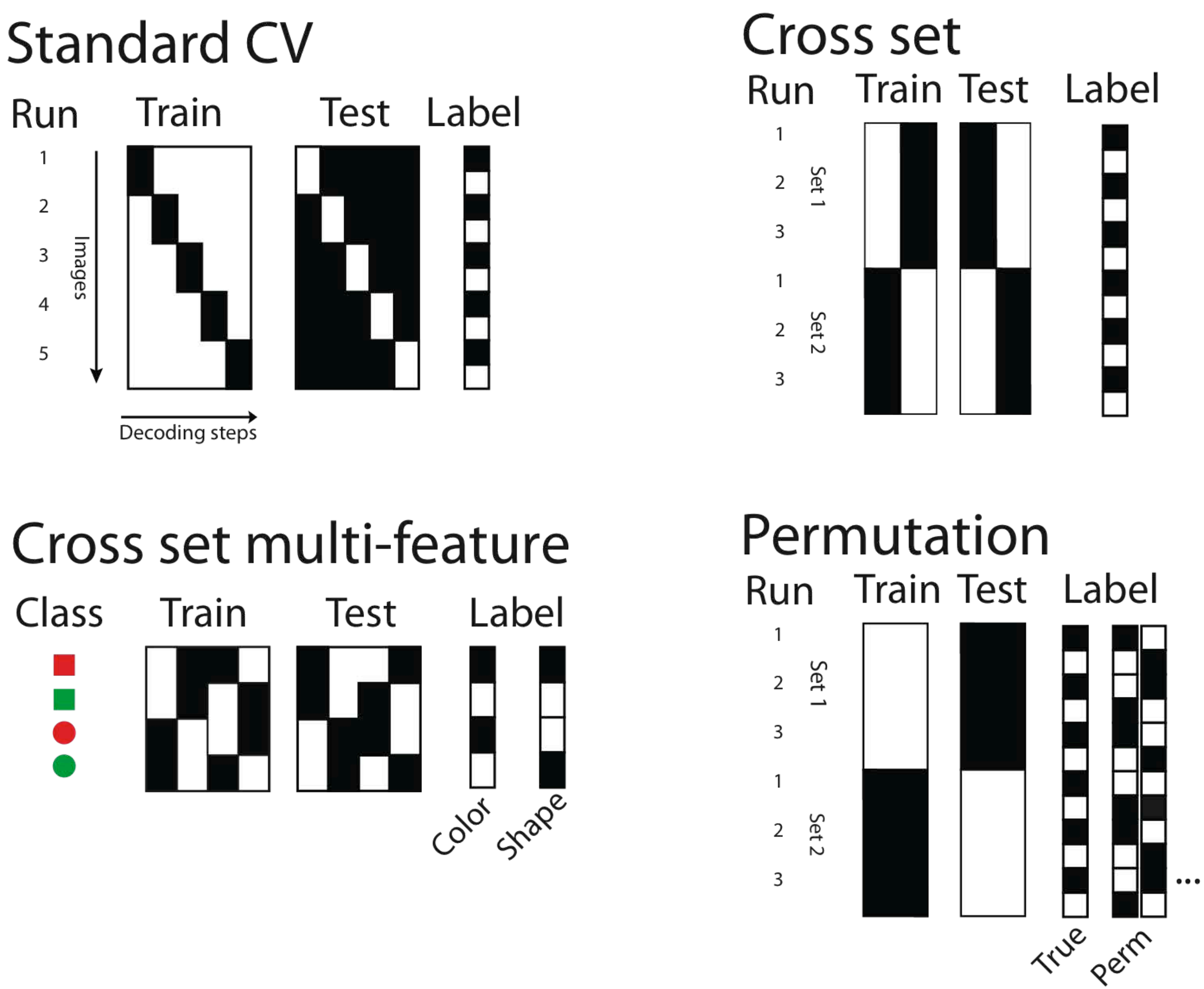
# A Standard Decoding Analysis



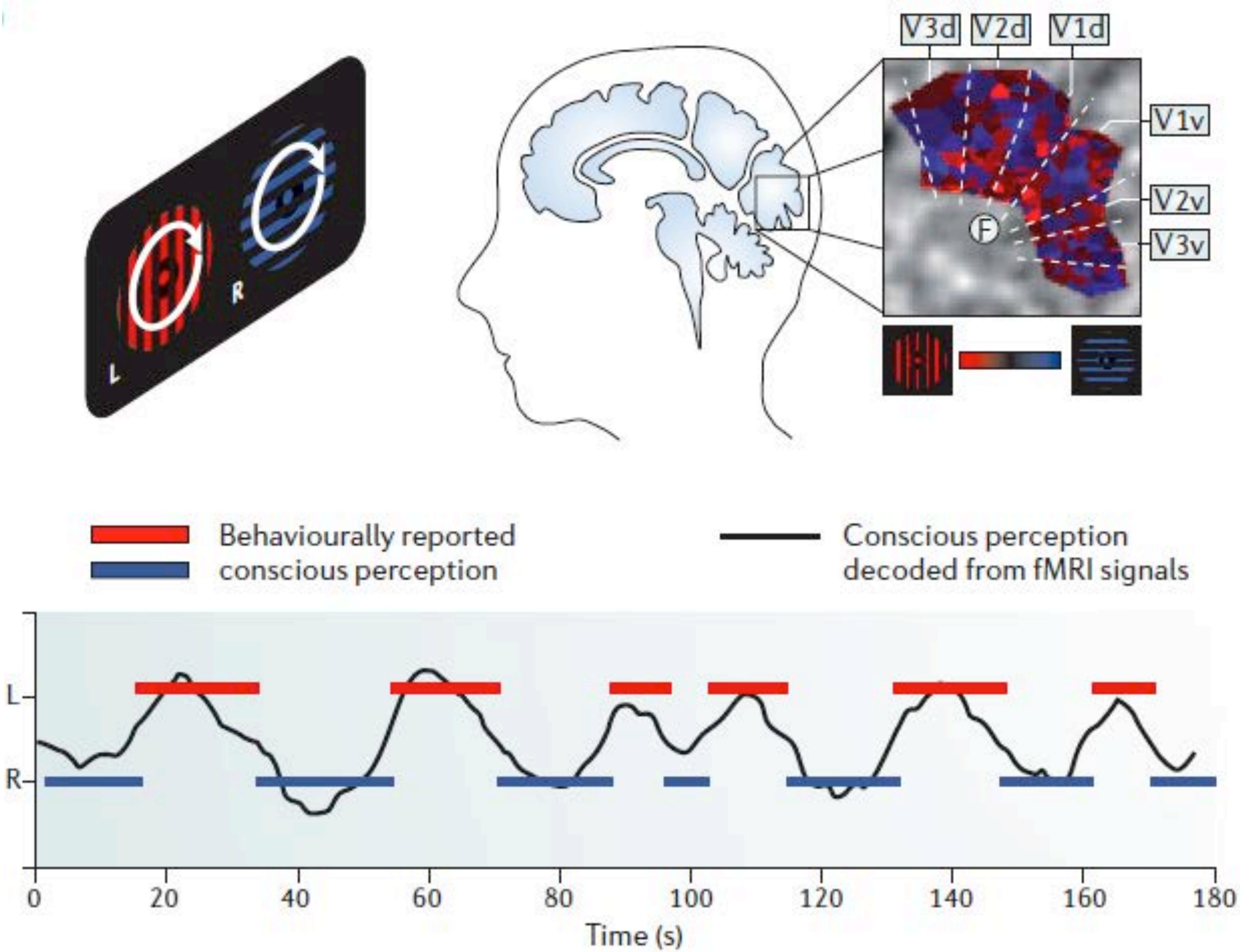


Input
Searchlight, whole brain, ROIs
Design Creation
Cross-validation, Cross-set, ...
Scaling
MinMax, z-transform, ...
Parameter Selection
Grid search, ...
Feature Selection
F-test, Recursive Feature Elimination, ...
Train & Test
SVM, SVR, Correlation classifier, ...
Predicted labels, Decision values
Transform Results
Decoding Accuracy, Correlation, AUC (Area under the ROC Curve), ...
Output
Information maps for SPM, mat-files
Whole brain
ROI
Searchlight
Designs
Standard CV
Run Train Test Label
Cross set multi-feature
Class Train Test Label
Color
Shape
Cross set
Run Train Test Label
Permutation
Run Train Test Label
True
Perm
Decoding Accuracy, Correlation, AUC, ...

# Different Decoding Designs

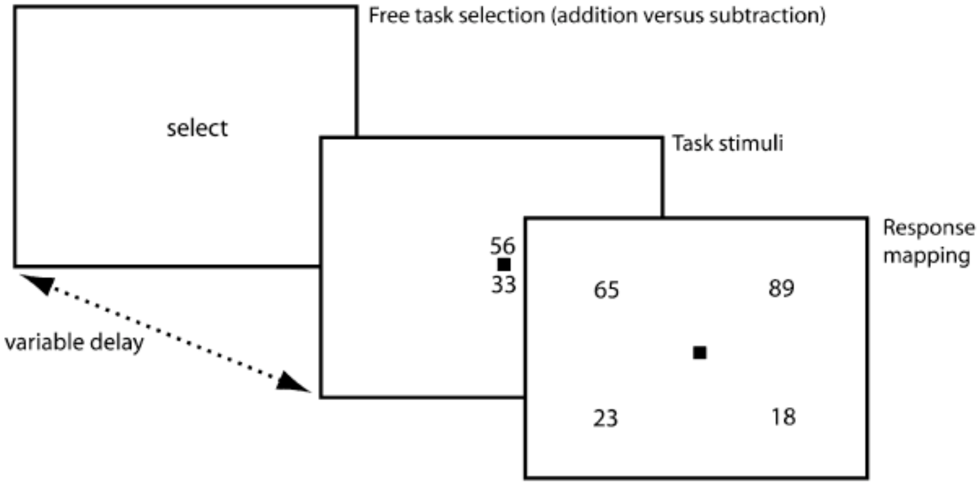


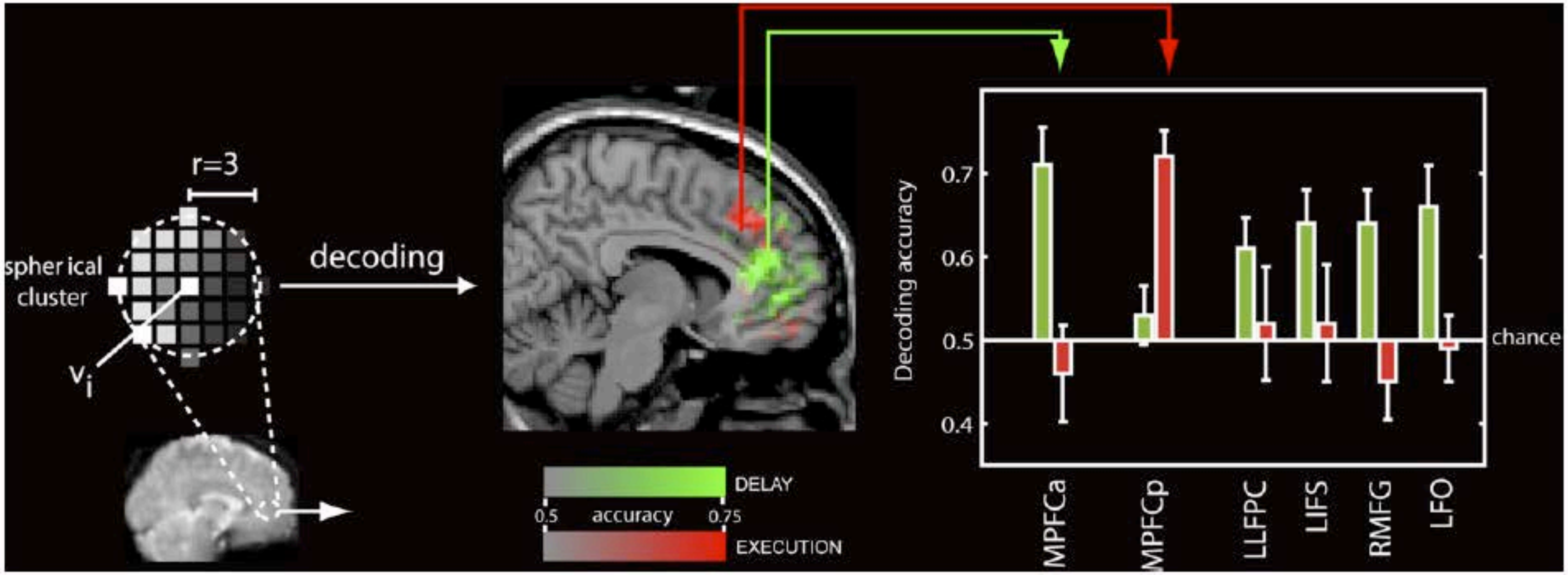
# Perceived Images




Haynes & Rees, NatRevNeuro, 2006

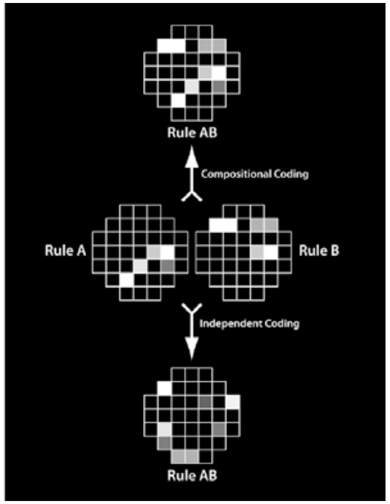
# Plus or minus

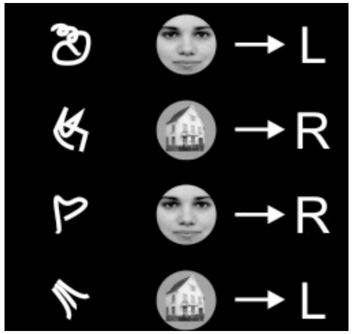


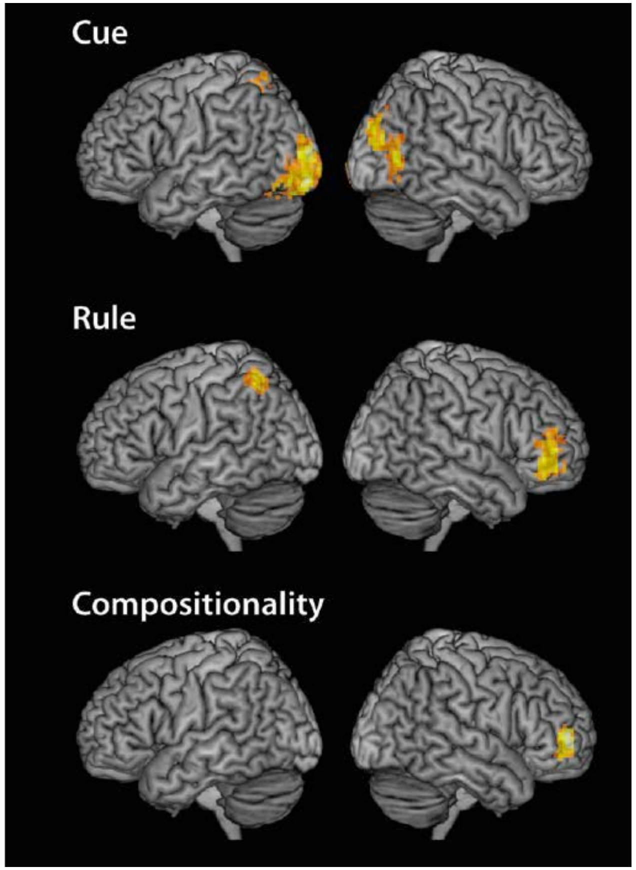



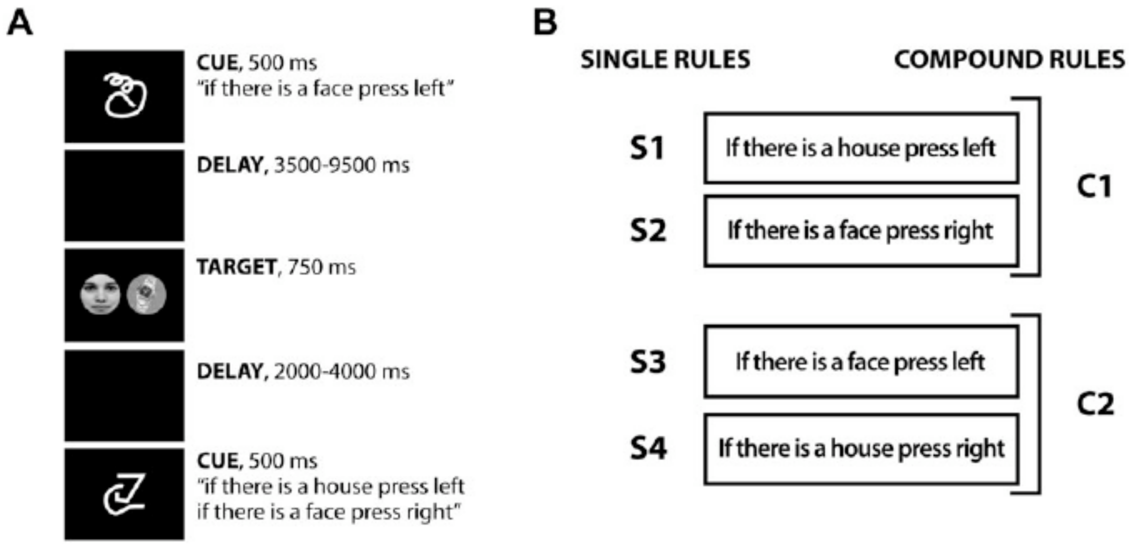

Haynes et al, CurrBio, 2007

# Are Rule Representations Compositional?




Reverberi, Görgen, Haynes, CerCor, 2011

# Some remarks

- Think about the design before the study
  - e.g. multiple repetitions of *exemplars* of a category
  - classification & regression possible
- Apply **Multiple-Comparison-Correction**!
  - e.g. in SPM: use FWE-correction on cluster-level
- Over-fitting **during training** make your analysis not wrong if you use an **independent test set**
  - e.g. Cross-validation, Cross-set-decoding
- Circular Analysis ('Voodoo')
  - No problem for Decoding with ROIs or Searchlight
  - An example for how a circular analysis could occur: Perform a Searchlight analysis first, the extract ROIs from this and apply them at the same positions
- Social brain responses in a dead fish
  - No problem, if you don't correct

# TUTORIALS

# Useful Software

For the tutorial

- The Toolbox
- MRIcron & MRIcroGL: Basic image viewer
- testdata

In general

- SPM: General fMRI & M/EEG analysis framework

# Toolbox ‘Installation’

- Extract the .zip archive to some folder
- Add this folder to your matlab path
  - addpath(‘c:\decoding_toolbox’);
- Check that it works calling
  - cfg = decoding_defaults

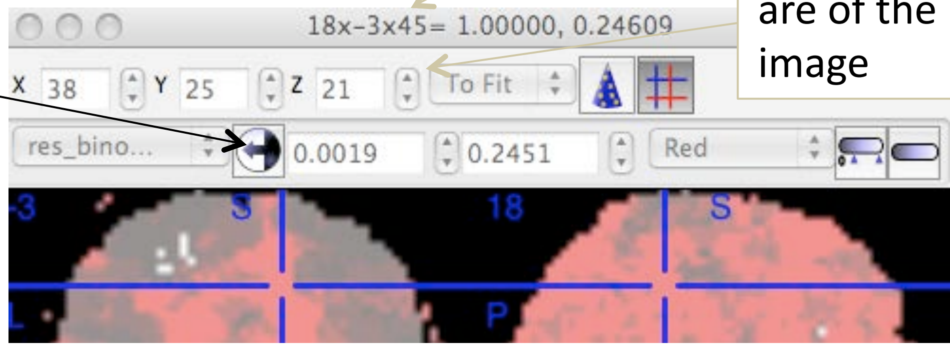
# Getting started: decoding_tutorial.m

- Open decoding_tutorial.m
- Fill in all required fields
  - labels: ‘left’ & ‘right’
    - left: left button, right hand index finger
    - right: right button, right hand index finger
- Run
  - should take about 3 mins

# Looking at your results with MRIcron & MRIcroGL

- General
  - Load brain mask (because result at the moment not normalized) with File -> Open
  - Load accuracy map as overlay
  - **CHANGE** the range to something meaningful
    - e.g. 60 – 100 for accuracies

Note: MNI-coord. at the top, XYZ-coord. are of the image




- MRIcroN shows values of overlays in the header (behind MNI coordinates)
- MRIcroGL has a nice real-time rendering

# General Explanation

- cfg
  - Important structure that contains all required information for the decoding
- decoding(cfg)
  - Function that does the work
  - Lot's of explanations about cfg settings at the top
- cfg = decoding_defaults()
  - All default settings + some instructions

# Example 2 – Toy Data Decoding

- Open demo\IK_demo2_toydata.m
- Run

- Tasks
  - Plot the hyperplane and the margins (i.e. get the model parameters as output measure)
  - Change SVM decoding parameter costs back to default (1) and see what happens to the boundary
  - Get the primal weights and plot them

# Example 3: How to add a result transformation

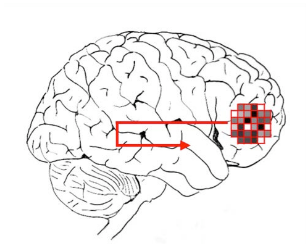
- Have a look at HOWTOEXTEND.txt in the decoding base folder and write a short result transformation that simply checks if the last prediction was correct

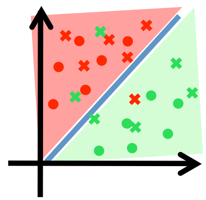
# The Decoding Toolbox
# How to get & Alternatives

- How to get The Decoding Toolbox?
  - Create a sourceforge account and e-mail us your login
- We provide on Sourceforge
  - Regular updates, Wiki, Room for questions
- If you have questions
  - e-mail us
- Current status
  - About 20 active users
  - Everything (seems) to work fine so far
  - Thus we are optimistic that it will do so in the future, too
- Alternatives
  - pyMVPA (Python, active community)
  - Princeton Decoding Toolbox (matlab, development stopped)

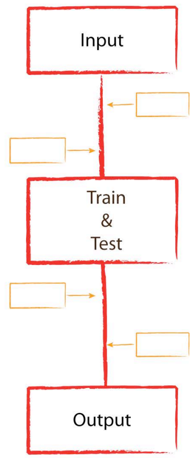
# Take Home

- fMRI detects new cars





- Decoding + Searchlight ‘increases’ the sensitivity of data analysis by looking for characteristic patterns




- The Decoding Toolbox makes decoding easy

# Thanks to

Martin Hebart

John-Dylan Haynes

Thank you for your attention

Questions?

Contact: Kai.goergen@bccn-berlin.de

# FURTHER THINGS TO PLAY WITH (IN CASE YOU LIKE)

# Things to play with

- Visualize the outcome of different 2d SVMs
  - RBF, polynomial, …
- Get the histogram for different output measures for random / separated data
  - re-run data simulation & analysis a couple of times & draw the outcome distribution
  - measures: decoding_accuracy, AUC, …
- Perform a multi-cross-set decoding analysis
  - simulate data
  - create design matrices (train, test, label)
  - run it & analyse it's result
- Implement a permutation test